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RE-EXAMINATION OF FAMA-FRENCH FACTOR INVESTING WITH CAUSAL INFERENCE METHODS

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Abstract

This study constructs a causal graph to analyze the relationship between Fama and French's (FF) Five Factors and stock returns, using causal discovery algorithms. Within the context of causal inference, a revised specification of the linear model is developed. The findings reveal that only HML (High Minus Low) and CMA (Conservative Minus Aggressive) factors validly serve as regressors for stock returns. These results call into question the validity of the FF model's current specifications.

1. Introduction

Factor investing can be defined as the investment approach that targets the exposure to measure factors that presumably explain differences in the performance of a set of securities. A five-factor model directed at capturing the size, value, profitability, and investment patterns in average returns that was developed by Fama French has been widely adopted in the industry. Specifically, the five-factor model is described as follows (Fama & French, 2015).

$$R_{it} - R_{Ft} = a_i + b_i(R_{Mt} - R_{Ft}) + s_i SMB_t + h_i HML_t + r_i RMW_t + c_i CMA_t + e_{it} \quad (1)$$

In this equation R_{it} is the return on security for period t, R_{Ft} is the risk-free return, R_{Mt} is the return on the value-weighted market portfolio, SMB_t is the return on a portfolio of small stocks minus the return on a portfolio of big stocks, HML_t is the difference between the returns on portfolios of high and low B/M stocks, RMW_t is the difference between return on portfolios of stocks with robust and weka profitability, CMA_t is the difference between the returns on portfolios of the stocks of low and high investment firms, which we call conservative and aggressive, and e_{it} is a zero-mean residual. In Fama French's procedure, securities in an investment universe are ranked according to one factor and two parallel operations are carried out on that ranking: (a) partition the investment universe into subsets delimited by quantiles, and compute the time series of average returns for each subset; and (b) compute the returns time series of a long-short portfolio, where top-ranked securities receive a positive weight and bottom-ranked securities receive a negative weight.

In Fama-French factor investing research, causal assumptions were never explicitly stated, however, the model specification they used – linear regression – was consistent with causal structure. The least-squares estimators from linear regression specification imply a directed flow of information because of the definition of the error terms, which is the portion of the stock returns that cannot be adjudicated to the factors. A researcher who chose a linear regression model has in mind a controlled experiment where X (factors) causes Y (stock return). Therefore, the objective of a factor model is not to predict Y, but to estimate the causal effect on Y. It is the researcher's responsibility to declare what particular causal graph informed the chosen specification such that the exogeneity assumption holds. The omission of causal graphs leads to logical inconsistency. Without a declared causal structure, the estimators lose causal meaning, and p-values merely convey the strength of association of unknown origins – causal and non-causal combined (Lopez de Prado et al., 2023).

In this paper, we re-examine the five-factor model proposed by Fama French by employing computational methods to identify causal relationships among five factors and stock return. A proper linear model is then proposed with an identified causal structure resulting from the algorithms. The paper is organized as follows. Section 2 delineates the data preparation process, including any necessary pre-processing steps undertaken prior to their integration into the computational algorithms. Section 3 presents an overview of three distinct categories of causal discovery algorithms. Section 4 elaborates on the robustness analysis conducted on the causal graph, which was derived from the aforementioned computational processes. Finally, Section 5 engages in a detailed discussion and proposes modifications to the factor model specifications, informed by insights gained from the causal graph analysis.

2. Data

In the interest of transparency and replicability, we detail our process for obtaining the data used throughout the paper below. Our goal was to replicate the initial conditions and datasets used in Fama and French's 2015 paper. For all data sources, we formed a time series that span July 1963 to December 2013, the same timeframe used by Fama and French (2015).

Monthly stock return data is obtained through Compustat and includes all stocks traded on the New York Stock Exchange, Nasdaq, and AMEX with share code 10 or 11. According to the Fama-French Model, this represents our dependent variable affected by their factors. In certain cases where a security undergoes changes (M&A, bankruptcy, new share classes, etc.) we see a change in the security and company codes. Therefore we choose PERMNO as our identifier as it reflects a specific security. As such, we partition our data into independent time series for each security, which will serve as the dependent variables when running the causal discovery algorithms. We also extract the five Fama-French factors (Mkt-Rf, SMB, HML, RMW, and CMA) from data publicly available on Kenneth French's website.

3. Methodology

Causal discovery is defined as the search for the structure of causal relationships by analyzing the statistical properties of observational evidence Spirtes et al., 2001. Numerous algorithms for the discovery of causal relations, represented as directed acyclic graphs (DAG) have been developed over the past three decades (Glymour et al. [2019]). The three major categories are: (a) constraint-based algorithms; (b) score-based algorithms; and (c) functional causal models. In this section, we describe the process of generating a causal graph for each category. The advantages and disadvantages are examined in detail.

3.1 Constraint-based algorithm

PC algorithm

The PC algorithm stands out as one of the earliest algorithms demonstrating consistency under i.i.d. sampling conditions, assuming the absence of latent confounders (Spirtes et al., [2001]). This algorithm offers a search framework wherein various statistical procedures can be integrated to determine conditional independence. The PC algorithm identifies the final causal graph by iteratively applying conditional independence tests to determine the absence of direct edges between variables, gradually refining the graph structure through a systematic search process.

If our choices about which variables are independent of each other hold up well with a ton of data (as we used the monthly data from March 1973 to December 2013), the PC algorithm is sure to sort out the accurate Markov Equivalence Class (MEC) in those conditions. Another similar algorithm is the SGS algorithm. They follow the same assumptions and the same consistency properties. However, the runtime of the PC algorithm is much faster, which performs fewer statistical tests. That is the reason why we choose to use it as a default algorithm for attempting causal discovery for the combination of the Fama-French factors and our macro factors.

3.2 Score-based algorithm

Two-Phase Greedy Search Algorithm

The Two-Phase Greedy Search Algorithm is a score-based algorithm originally proposed by Meek in 1997. David Maxwell Chickering provides a new implementation of the search space by using equivalence classes as stated in 2002. Bayesian scoring criterion is applied during the implementation. Below shows a parameterized Bayesian network over a joint distribution U that factors according to structure G as follows:

$$P_B(X_1 = x_1, \ldots\ldots, X_n = x_n) = \prod_{i=1}^{n} p(X_i = x_i \mid Pa_i^G = pa_i^G, \theta_i)$$

Where :

Pa_i^G: The set of parents of node x_i in G

The process of the algorithm can be summarized as follows:

1. We start with an equivalence class corresponding to no dependencies.
2. As the first phase of the greedy search algorithm, we add dependencies by considering all possible single-edge additions that can be made to all DAGs in the current equivalence class until the algorithm stops at its local maximum.
3. We then apply the second phase of the greedy algorithm that considers all possible single-edge deletions that can be made to all DAGs in the current equivalence class.

4. The algorithm is terminated at the local maximum identified in the second phase.

This approach efficiently identifies the perfect map over a given generative distribution if that map is DAG. Therefore, this approach does not support feedback loops or bi-directed edges (unmeasured confounders).

3.3 Functional Model

In addition to the constraint-based algorithm like PC-algo and score-based algorithm like GES algo that we introduced before, we also considered another algorithm called functional causal models.

Functional causal models represent the effect Y as a function of the direct causes X and some unmeasurable noise, and the noise is assumed to be independent of the causes X. Compared to constraint-based causal discovery, causal discovery based on functional causal models is able to identify the whole causal model under appropriate assumptions. In addition, according to previous literature, traditional causal discovery algorithms like constraint-based or score-based algorithms, produce a Markov equivalent class of the causal models, in which some causal directions may be undetermined. On the other hand, a functional causal model, which expresses each variable as a function of its direct causes and the independent disturbance, if well specified, can explain the data-generating process and help find all causal relations among the variables.

There are mainly three types of functional causal models: Linear, non-Gaussian, and acyclic model (LiNGAM [Shimizu et al. 2006]), nonlinear additive noise model ([Hoyer et al. 2009]), and post-nonlinear causal model (PNL [Zhang and Hyvärinen 2012]). In LiNGAM, the function is linear, and at most one of the noise and cause is Gaussian: $x_i = \sum_{k(j)<k(i)} b_{ij} x_j + e_i + c_i$, where e_i is the noise term, c_i is an optional constant term. In the nonlinear additive noise model, the function is nonlinear with additive noise: $x_i = f_i(x_{pa(i)}) + n_i$, where $pa(i)$ is the direct causes of x_i, f_i is an arbitrary function (possibly different for each i, $x_{pa(i)}$ is a vector containing the elements x_j such that there is an edge from j to i in the DAG, and n_i the noise variables that are jointly independent. In PNL causal model, the function is further generated by a post-nonlinear transformation on the nonlinear effect of the cause X plus the noise: $x_i = f_{i,2}(f_{i,1}(pa_i) + e_i),\ i = 1, \ldots, n$, where pa_i is the direct cause of x_i, $f_{i,1}$ is the nonlinear effect of the causes, e_i is an independent disturbance, $f_{i,2}$ is an invertible post-nonlinear distortion in variable x_i.

To implement the functional causal model in Python, we used a package called *cdt.causality.graph.CAM*. It is based on causal additive models, which rely on fitting Gaussian Processes on data, while considering all noise additives and additive contributions of variables. The *cdt* package can detect the causality among variables pairwisely and plot the causal graph.

4. Robustness Analysis

Unveiling the causal relationships between the Fama-French factors and stock returns involves a robustness analysis. The goal is to evaluate the stability and strength of the causal links across different conditions. Identifying changes in the output causal graph provides valuable insights into the model's adaptability. For all the algorithms that we chose, the resilience of the final causal graph to data loss or alterations is explored by evaluating the returns of one stock at a time, assessing whether the network edges remain unaffected or undergo significant changes. For each causal graph, we record the appearance of all the directed edges, which sum up to be the total number of occurrences for each causal relation. These numbers, divided by the total number of stocks with available data, produce the weights.

$$w_{(i,j)} = \frac{\#\, Occurrence\ of\ edge\ (i,j)}{\#\, Total\ stocks}$$

A higher weight of an edge explains good stability and persistence across diverse scenarios. On the other hand, a lower weight indicates less stability and less generalization of the causal relationship the edge is representing.

Comprising the five Fama-French factors and an additional variable representing the observed stock return, a complete graph encompasses a total of *6C2 = 15* undirected edges. Since every causal link is directional and should be considered separately, the total number of directed edges we are interested in amounts to 30. Sorting the edges by their corresponding weights that we obtained using the described procedure, we produce a simple skeleton by greedily adding them one by one until the first ambiguity is observed. Specifically, we started from an empty graph and added the directed edges from larger weights to smaller weights, until the second cycle occurs. The only cycle we allowed in this skeleton is the cycle between HML and CMA. The reason behind this is that, these two factors appear to have a relatively high association and the causal relationships in both directions are strong. If we terminate early as soon as this first cycle appears, then the skeleton would be too sparse where not enough links are going in or out from the stock return variable. On the other hand, if we allow too many cycles to exist, the skeleton would become too dense where the later-added edges are redundant and not representative of the causal paths we are trying to identify.

	edge	score	weight	weight(%)
0	(HML, RMW)	5404	0.559362	55.94
1	(CMA, Mkt-RF)	4901	0.507297	50.73
2	(HML, SMB)	4839	0.500880	50.09
3	(HML, Mkt-RF)	4667	0.483076	48.31
4	(CMA, HML)	4663	0.482662	48.27
5	(Return, Mkt-RF)	4651	0.481420	48.14
6	(HML, CMA)	4639	0.480178	48.02
7	(SMB, Mkt-RF)	4439	0.459476	45.95
8	(RMW, Mkt-RF)	4296	0.444674	44.47
9	(CMA, RMW)	4244	0.439292	43.93
10	(SMB, RMW)	4136	0.428113	42.81
11	(HML, Return)	3930	0.406790	40.68
12	(CMA, SMB)	3871	0.400683	40.07
13	(Return, SMB)	3821	0.395508	39.55

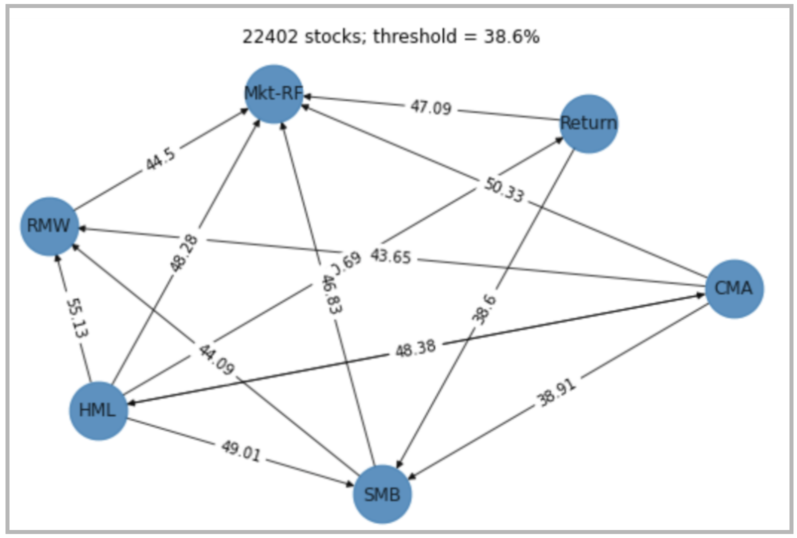


Figure 1A (left) 13 out of the 30 directed edges are kept by a threshold of 38.6%; Figure 1B (right) the resulted graph with the 13 edges

From the skeleton we produce using the functional model, 13 out of the 30 directed edges are kept by a threshold of 38.6% applied to the weights. The skeleton is acyclic should the CMA and HML cycle be absent. In addition, HML is the only direct cause of the stock return variable.

We apply the same process and terminating conditions to using PC and GES, however, the resulting skeletons are much sparser. For GES (result shown on the left), the outcome variable has only one causal link that's going out, and for PC (result shown on the right), the result is less informative – the presence of {HML, CMA} cycle dwarfs the stability of all other causal links. Therefore, we go further by using the skeleton produced by the functional model alone, to identify the causal paths and examine the factor specification of Fama and French's choice.

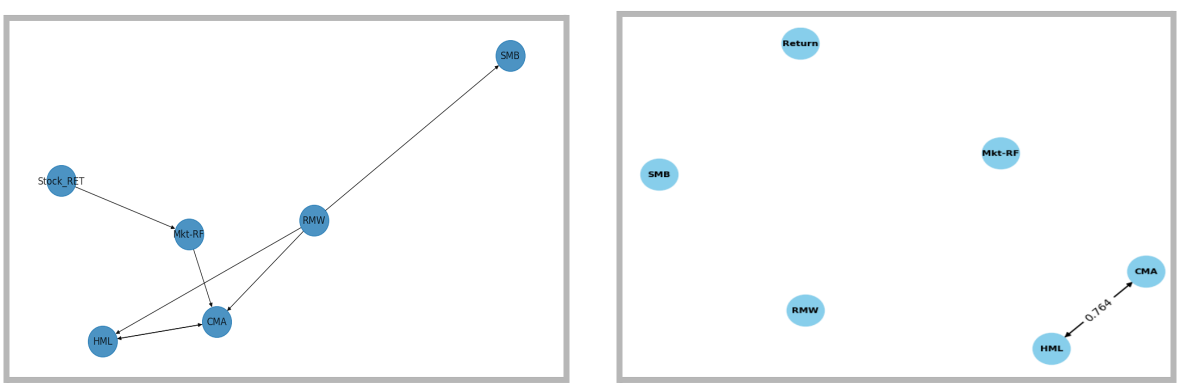
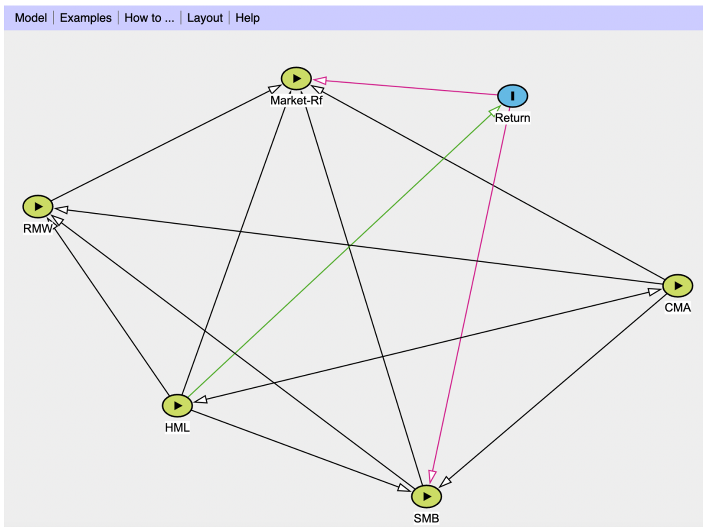


Figure 2A (left) the resulted causal graph produced by GES; Figure 2B (right) the resulted causal graph produced by PC

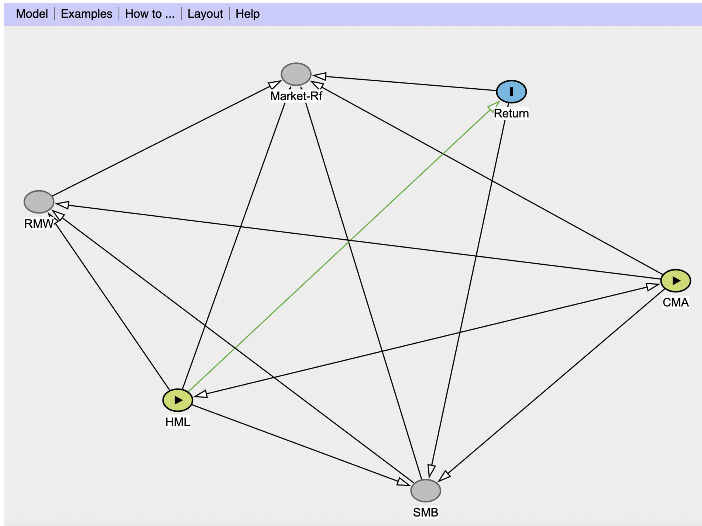
5. Discussion of Factor Specification

We utilized an online tool named DAGitty (Texter et al., 2016) to effectively identify the causal paths and backdoors given the above skeletons. The tool requires a directed acyclic graph as input, as well as the selections of exposure and outcome variables, which in our study correspond to the Fama French factors and the stock returns, respectively.



Exposure variable(s): all five Fama French factors   Outcome: stock return
Green path(s): causal   Pink path(s): biasing
Biasing paths are open. No adjustment set is found.

When all five factors are selected to be the exposure, as what Fama French regression indicates, we observe two open biasing paths, and no adjustment set is found, meaning there's no way to control the variables such that these biasing paths can be blocked. This implied that the factor specification proposed by Fama and French is problematic. To provide an alternative, we iteratively deselect Fama French factors to be the exposure variable in an attempt to find the largest set of factors, under the exposure of which there exist no open biasing paths.



Exposure variable(s): {HML, CMA} Outcome: stock return
Green path(s): causal Pink path(s): biasing
No open biasing paths.

Selecting {HML, and CMA}, we obtain a valid graph where one causal path is identified from HML to the stock return. This prompts the assumption of modifying Fama and French's linear regression by using HML and CMA alone as the causal variables.

A further observation worth discussing is that the causal graph displays directed edges from HML to every other node, so this suggests a causal relationship between not only HML and returns but also HML and each of the other factors. Interestingly, Fama and French (2015) describe HML as a redundant factor since regressing HML returns on the other four factors yields an intercept near zero. In other words, they claim the average HML return is captured by the exposures of HML to other factors, explaining why a four-factor model without HML does not significantly hinder the description of average returns compared to the five-factor model. However, our results suggest that HML may be exogenous rather than redundant, and this relationship observed by Fama and French may be due to the causal links between the factors described in the causal graph.

6. Conclusions

In our study, the application of causal discovery to the Fama and French (FF) Five Factors reveals flaws in the current model's specification. There are biased pathways between {Market-Rf, stock returns} and {SMB, stock returns} in the causal graph, indicating substantial estimator bias by controlling for wrong variables. This bias leads to misleading results, including false positives and negatives.

By revising the exogenous variables, aiming to obtain valid causal links, it identifies {HML and CMA} as the most valid set of factors, with HML having a direct causal influence on stock returns. The resulting causal graph also suggests a causal linkage of HML with the other four factors, contradicting Fama and French's assertion of HML's redundancy. This finding substantiates our conclusion that HML is a critical causal factor.

In light of these observations and results, we recommend future research to concentrate on quantifying the coefficients for HML and CMA within a redefined linear model.

References

Fama, E. F., & French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, *116*(1), 1–22. https://doi.org/10.1016/j.jfineco.2014.10.010

Glymour, C., K. Zhang, and P. Spirtes (2019): “Review of causal discovery methods based on graphical models.” Frontiers in Genetics, Vol. 10, p. 524. Available in https://www.frontiersin.org/articles/10.3389/fgene.2019.00524/full

Hoyer, P., Janzing, D., Mooij, J. M., Peters, J., & Schölkopf, B. (2008). Nonlinear causal discovery with additive noise models. Advances in neural information processing systems, 21.

Lopes de Prado, M. M. (2023). 6.1. Causal Content, In Causal factor investing: Can factor investing become scientific? essay, Cambridge University Press.

Shimizu, S., Hoyer, P. O., Hyvärinen, A., Kerminen, A., & Jordan, M. (2006). A linear non-Gaussian acyclic model for causal discovery. Journal of Machine Learning Research, 7(10).

Spirtes, P., C. Glymour, and R. Scheines (2001): Causation, Prediction, and Search. Cambridge, MA: MIT Press, 2nd edition.

Textor, J., van der Zander, B., Gilthorpe, M. K., Liskiewicz, M., & Ellison, G. T. H. (2016). Robust causal inference using directed acyclic graphs: the R package 'dagitty'. International Journal of Epidemiology, 45(6), 1887-1894.

Zhang, K., & Hyvarinen, A. (2012). On the identifiability of the post-nonlinear causal model. arXiv preprint arXiv:1205.2599.

Appendix

I. *Data*

We aimed to include other factors that we believed may contribute to the dynamics they assert in their work alongside the five Fama-French factors. The factors that we decided to include were the TERM and DEF factors specified in Fama and French (1993) as well as GDP and inflation.

The TERM factor, as they describe, is a proxy for unexpected changes in interest rates, so our alternate proxy for this phenomenon is the difference between 3-month and 10-year US Treasury yields. Meanwhile, DEF is described as a proxy for shifts in economic conditions that change the likelihood of default, so we utilize the difference between the LUACTRUU Corporate Bond Index yield and the 10-year UST yield. One pitfall of this DEF factor is that the corporate bond data is only available as of February 1973, which is the oldest corporate bond portfolio data we could find but does not align with the desired 1963-2013 timeframe.

To introduce GDP and Inflation factors, we chose two indices to act as proxies. In the case of GDP we use the Brave-Butters-Kelly index, extracted from the Federal Reserve Economic Data (FRED) database, as a proxy because GDP is declared quarterly. Our proxy instead represents annualized monthly GDP growth which is indexed to official quarterly estimates of real GDP growth.

To include inflation as a factor, we choose the Consumer Price Index for All Urban Consumers: All Items in the U.S. City Average series (CPIAUCSL) from FRED, and more specifically the annualized month-over-month return of this index.

II. *Other model results*

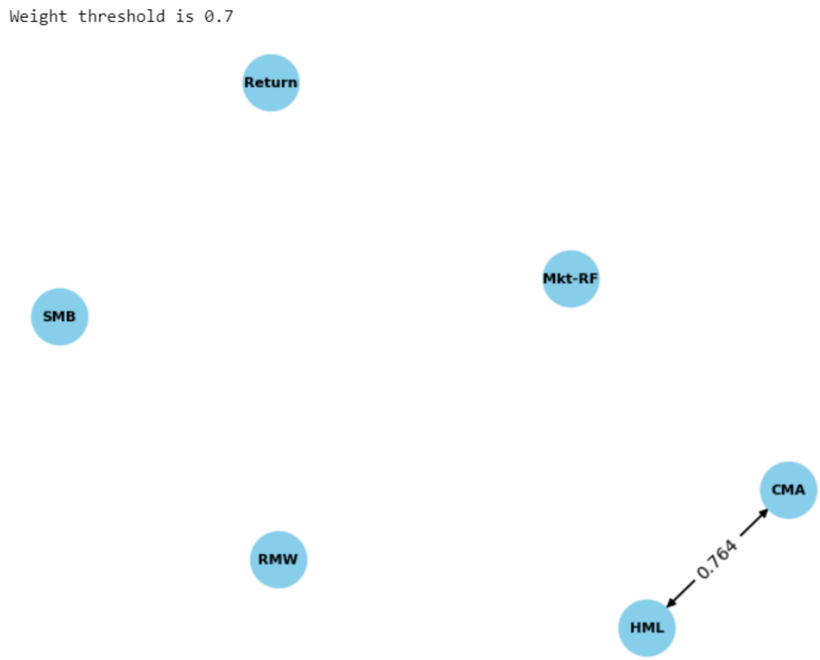
A. PC algorithm Result

To test whether the PC algorithm can yield similar causal relationships in different data samples, we conducted a Robustness Analysis. Among approximately 23,000 stock return data points, we randomly selected 10,000 stocks, treated their returns as a new factor, and reran the PC algorithm. Here are the 5 edges with the highest occurrence.

<u>Summary of results with FF factors only</u>

Starting node	**Ending node**	**Count**	**Frequency**
CMA	HML	7645	0.764
HML	CMA	6882	0.688

CMA	Market - Risk free	4233	0.423
Return	Market - Risk free	3862	0.386
Market - Risk free	Return	3643	0.364
…	…	…	…



<u>Summary of results with macro factors</u>

Starting node	**Ending node**	**Count**	**Frequency**
CMA	HML	4290	0.715
HML	CMA	3955	0.659
CMA	Market - Risk free	2390	0.385
TERM	CPI	2178	0.363
GDP	DEF	2154	0.359
…	…	…	…

If we sequentially add edges from the highest occurrence to the lowest until a cycle appears, we can quickly obtain a cycle where CMA and HML factors point to each other. This result is somewhat unexpected because we cannot extract stable factors that are influential to returns from the PC algorithm's results. However, this also acknowledges that the PC algorithm is highly informative. The PC algorithm considers both direct and indirect dependencies between variables, aiming to uncover causal relationships. In other words, the PC algorithm provides a more comprehensive representation of potential causal relationships, leading to the cycle we have.

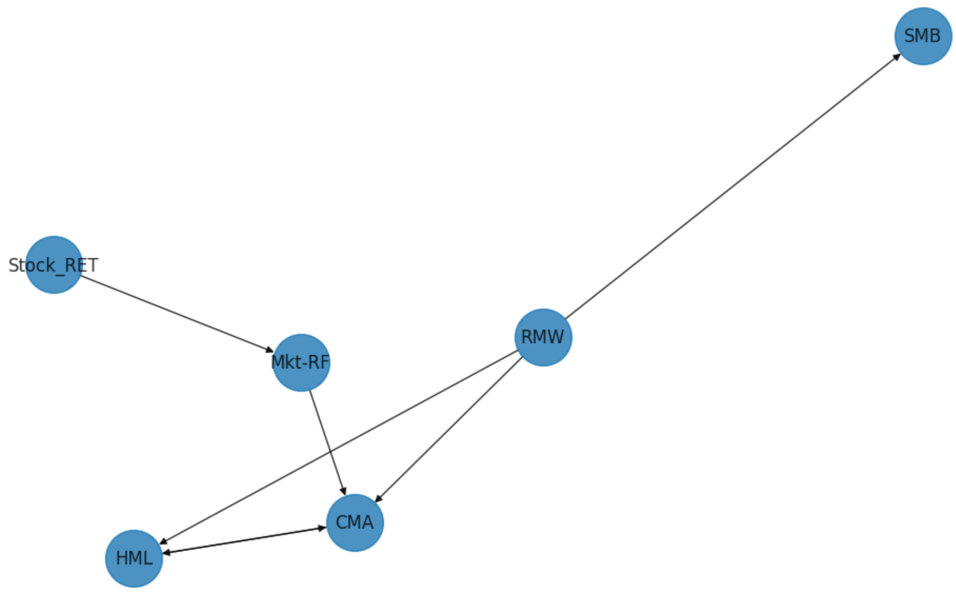
B. GES algorithm Result

Using the similar methods aforementioned we generated the causal graph using both FF factors and macroeconomic factors. The plots are shown as follows.

<u>Summary of results with FF factors only</u>

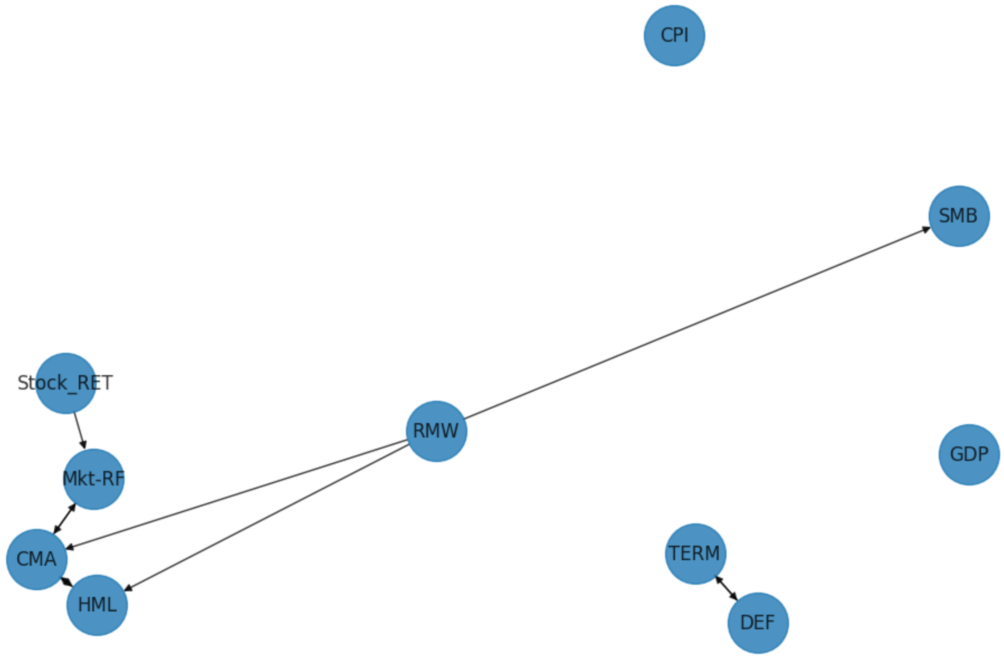
Starting node	**Ending node**	**Frequency**
CMA	HML	0.6821
HML	CMA	0.5394
Market - Risk free	CMA	0.5116
RMW	HML	0.5014
RMW	SMB	0.4259

RMW	CMA	0.402435
Stock Return	Market - Risk free	0.398174
Market - Risk free	Stock Return	0.388435



Summary of results with macro factors

Starting node	**Ending node**	**Frequency**
CMA	HML	0.6264
HML	CMA	0.527545
Market - Risk free	CMA	0.464182
RMW	HML	0.457636
TERM	DEF	0.444273
RMW	SMB	0.417364
RMW	CMA	0.388318
DEF	TERM	0.380091


CPI
SMB
Stock_RET
RMW
GDP
Mkt-RF
TERM
CMA
HML
DEF